

A Data-Driven Study of Residential Location Preferences and Urban Living Patterns Among Different Demographic Groups

1. Project Overview and Objective

- The main objective is to analyze residential location preferences using data pre-processing and analysis techniques in Python, along with an interactive Power BI dashboard, to identify the key financial, demographic, safety, commute, and facility-related factors influencing people's choice of residential areas.
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2. Data Sources

- **Source Description and Timeline:** Survey data collected for an academic project and 2024.
 - **Domain:** Real Estate and Residential Housing Analytics
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3. Problem Statement

- To study residential location preferences and understand why people choose one residential area over another based on factors such as rent, safety, commute, transportation, and nearby facilities.
 - To identify how demographic and financial factors such as age, occupation, household type, and annual income are associated with residential area selection.
 - To analyze the dataset because residential location is a common real-world decision influenced by multiple factors, making it suitable for identifying meaningful patterns and relationships through data analysis.
 - To determine which factors have the strongest influence on residential preferences and compare how these factors vary across different areas and groups of people.
 - To derive meaningful insights from the data that can help understand residential preferences and the characteristics associated with different location choices..
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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Person_ID	Object	Unique identifier for each person
Occupation	Object	Employment category of the respondent
Area_Selected	Object	Preferred residential area
Monthly_Rent	Float	Monthly housing rent
Distance_To_Work	Float	Distance between home and workplace
Public_Transport	Object	Availability or quality of public transport
Safety_Rating	Float	Perceived neighborhood safety score
Nearby_Hospitals	Object	Access to hospitals nearby
Nearby_Schools	Object	Access to schools nearby
Shopping_Access	Object	Availability of shopping facilities
Noise_Level	Object	Perceived neighborhood noise level
Internet_Availability	Object	Availability or quality of internet service
Main_Reason	Object	Primary reason for choosing the area
Household_Type	Object	Household composition, such as Single, Couple, or Family
Age	Integer	Age of the respondent
Annual_Income	Float	Annual income of the respondent
Monthly_Income	Float	Monthly income calculated from Annual Income by dividing it by 12.
Rent_Income_Ratio	Float	Ratio of Monthly Rent to Monthly Income showing the proportion of income spent on rent.

5. Tools & Technologies

- **Python:** Used Python, Pandas, NumPy, Matplotlib, Seaborn for data preprocessing, cleaning, and analysis
- **Power BI:** Visualization, and interactive dashboard creation.

6. Data Pre-Processing (Python / Pandas)

Tasks Performed:

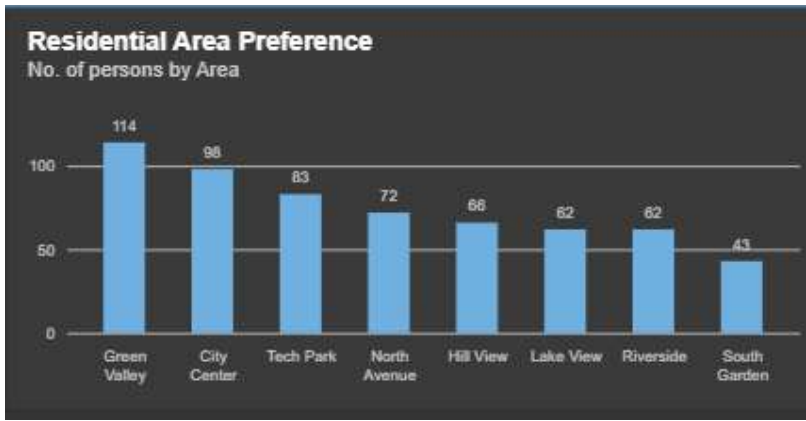
- **Data Loading & Initial Inspection:** Loaded the dataset using Pandas and examined its structure, number of rows and columns, data types, and basic statistics.
 - **Column Name Standardization:** Removed unnecessary spaces from column names to maintain consistent naming.
 - **Duplicate Handling:** Checked for duplicate records and duplicate Person_ID values. Duplicate Person_ID records were removed while retaining the first occurrence, resulting in **600 unique respondents**.
 - **Missing Value Treatment:** Identified missing values in numerical and categorical columns.
 - Categorical missing values were filled using the **mode**.
 - Monthly_Rent, Distance_To_Work, and Safety_Rating were imputed using **area-wise median values**.
 - Age and Annual_Income were imputed using their **overall median values**.
 - **Categorical Data Standardization:** Standardized categorical values by removing extra spaces and applying consistent capitalization.
 - **Outlier Detection & Treatment:** Used the **Interquartile Range (IQR) method** to identify unusual numerical observations. Invalid/extreme values were treated using the respective column's median where appropriate.
 - **Invalid Value Correction:** Negative values in fields such as rent, distance, and annual income were identified during data validation and corrected where necessary.
 - **Skewness Analysis:** Checked the skewness of numerical variables. The variables did not show severe skewness, so no major transformation such as logarithmic transformation was required.
 - **Feature Engineering:** Created: $\text{Monthly_Income} = \text{Annual Income} / 12$ and $\text{Rent_Income_Ratio} = \text{Monthly Rent} / \text{Monthly Income}$
 - **Data Validation:** Rechecked missing values, duplicates, data types, and numerical ranges after preprocessing to ensure the dataset was ready for analysis.
 - **Clean Dataset Creation:** Exported the final cleaned dataset as a CSV file and used it for further analysis and Power BI visualization.
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7. Visualizations(Power BI)

Visualisations	Purpose
Residential Area Preference – Column Chart	To compare the number of respondents selecting each residential area and identify the most preferred areas.
Average Monthly Rent by Residential Area – Bar Chart	To compare the average rental cost across different residential areas.
Average Safety Rating by Residential Area – Area Chart	To compare perceived safety levels across residential areas.
Main Reasons for Residential Area Selection – Donut Chart	To understand the major factors influencing respondents' residential area selection.
Average Rent-to-Income Ratio (%) by Area – Line Chart	To compare rental affordability/burden across different residential areas.
Public Transport Usage by Residential Area – 100% Stacked Column Chart	To examine the availability/usage of public transportation across different areas.
Monthly Income vs Monthly Rent – Scatter Plot	To examine the relationship between respondents' income and their monthly rent.
Slicers – Area, Household Type and Occupation	To allow interactive filtering and comparison of the dashboard based on different respondent groups.

Example Visualization and Interpretation

Example: Residential Area Preference



Purpose:

The column chart was created to compare the number of respondents who selected each residential area. This helps identify which residential areas are more preferred among the respondents.

Interpretation:

The chart shows that the number of respondents varies across the different residential areas. Some areas have a higher number of selections, indicating relatively greater residential preference, while other areas have fewer selections. This provides an overall understanding of the distribution of residential preferences among the respondents.

Note:

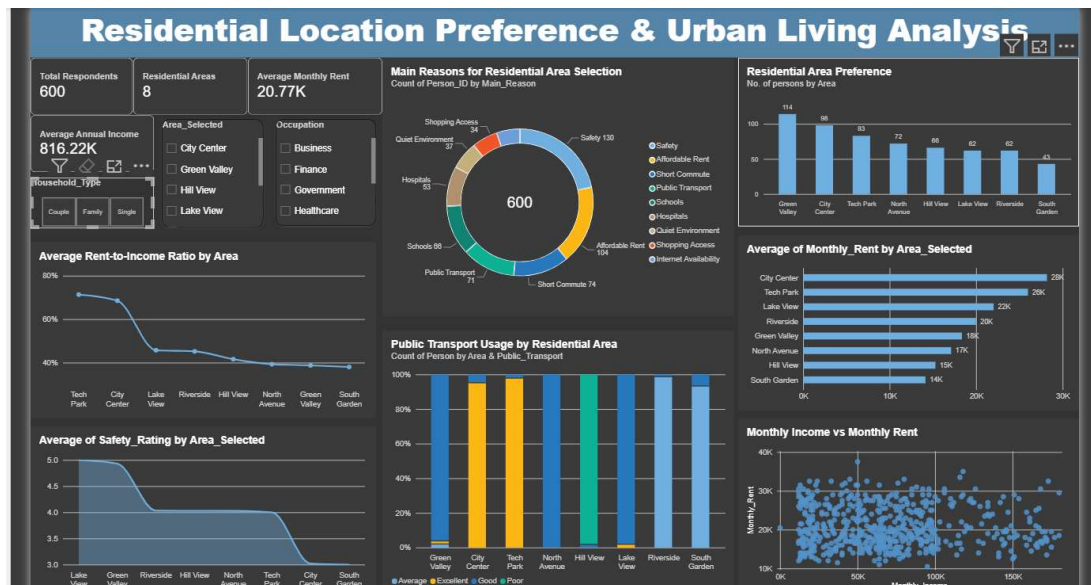
Similar visualizations were created for other important variables such as monthly rent, safety rating, main reasons for area selection, rent-to-income ratio, public transport, and income versus rent. Each visualization was selected according to the analytical objective and was accompanied by an appropriate interpretation to identify meaningful patterns and relationships.

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- The dashboard includes **KPI cards** for total respondents, residential areas, average monthly rent, and average annual income. Different visualizations were created, including **column charts, bar charts, donut charts, line charts, a 100% stacked column chart, and a scatter plot** to analyze residential area preference, rent, safety, main reasons, rent-to-income ratio, public transport ratings, and the relationship between income and rent.
- The dashboard shows that **Green Valley has the highest number of selections (114)**, while **South Garden has the lowest (43)**. **Safety** is the most common reason for selecting a residential area, followed by **affordable rent**. Average rent is highest in **City Center (approximately ₹28K)** and lowest in **South Garden (approximately ₹14K)**. The rent-to-income ratio provides an understanding of rental affordability across areas, while the income-versus-rent scatter plot helps examine their relationship.
- Interactive **slicers for Residential Area, Household Type, and Occupation** were also added to allow users to filter and compare the results based on different groups.

SCREENSHOT OF DASHBOARD /REPORT



9. Insights & Conclusions

Key Findings

- Green Valley has the highest residential preference with 114 respondents, while South Garden has the lowest with 43 respondents.
- Safety is the most common reason for selecting a residential area, followed by affordable rent.
- City Center has the highest average monthly rent, while South Garden has the lowest.
- Distance to work and monthly rent show a moderate negative relationship, indicating that areas farther from workplaces tend to have lower rents.
- The rent-to-income ratio varies across areas, highlighting differences in housing affordability.
- Monthly income and monthly rent show very little linear relationship, suggesting that income alone does not explain rental differences.
- Public transport ratings and safety levels also vary across residential areas.

10. Conclusions

The study concludes that people choose residential areas based on different factors such as safety, rent, distance to work, transportation, and nearby facilities. Safety and affordable rent are the main factors found in the analysis. Rent also differs across residential areas, while income alone does not have a strong relationship with rent. Overall, the study shows that residential decisions depend on a combination of different factors, rather than just one factor.
